Programming Spec: Steam Research and Insights Tool
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General description of the project

The Steam Research and Insights Tool is designed to support User Research and Insights teams by collecting and analyzing player conversations from Steam community forums, discussion boards, and reviews related to specific video games or video game genres. The platform serves as a secondary research solution that enables researchers and player insights teams to compare large-scale player feedback without manually reviewing thousands of individual posts against smaller internal studies.

The system collects and organizes user-generated content from filtered Steam communities, allowing teams to identify trends and themes related to player sentiment, usability concerns, reactions to new gameplay mechanics, and competitive comparisons within a genre.

Other relevant stakeholders can use this tool to search for specific games, franchises, or genres and filter results by date range to examine community discussions during particular time periods, such as game launches, major updates, downloadable content releases, or seasonal events. The tool organizes the data into relevant categories and themes that stakeholders can use to compare with internal data for actionable feedback.

By providing centralized access to player feedback and community sentiment, the tool enables research and marketing teams to better understand player experiences on a larger scale that have been identified in smaller instances, identify emerging issues, monitor market competition, and support data-driven decision-making throughout the game development lifecycle.

Task Vignettes

Task 1: Search by Video Game Title

Task 1 User Activity:

1. The User Research team member logs into the web application.
2. The user enters the name of a video game (e.g., Call of Duty) into the search bar.
3. The user selects the desired Steam community discussion boards and review sources associated with the game.

4. The user clicks the "Search" button to begin data collection.
5. The system retrieves relevant discussion posts, comments, and reviews related to the selected game.
6. The system processes the collected data and prepares it for analysis.

Outcome: The research and insights team receives a broad overview of player discussions of a particular topic related to the selected game.

Task 2: Filter by Date Range

Task 2 User Activity:

1. The user selects a start date and end date using the date range filter.
2. The user applies the filter to limit results to discussions posted during the specified timeframe.
3. The system re-runs the query using only posts and comments that fall within the selected date range.
4. The dashboard updates to display findings specific to that period.
5. The user reviews trends that may be associated with game launches, updates, patches, events, or seasonal content.

Outcome: The user can analyze player sentiment and feedback during a specific timeframe and identify changes in community response over time.

Task 3: Search for a Particular Mechanic in Call of Duty

Task 3 User Activity:

1. The user searches for "Call of Duty" in the game search field.
2. The user enters a specific mechanic or keyword (e.g., skill-based matchmaking, weapon balancing, movement system, prestige progression) into a secondary search field.
3. The user optionally applies a date range filter to focus on recent discussions.
4. The system searches Steam discussions and reviews for mentions of the selected mechanic.
5. The system identifies and groups discussions related to the mechanic into categories such as:
    - Positive Sentiment
    - Negative Sentiment
    - Usability Concerns
    - Competitive Comparisons
6. The dashboard displays:
    - Overall sentiment toward the mechanic
    - Frequently discussed themes

    - Representative player quotes
    - Volume of discussion over time
7. The researcher reviews the findings and compares them with internal playtest data to determine whether observed player behaviors and opinions are isolated or representative of the broader player community.

Outcome: The user gains a focused understanding of player perceptions and experiences related to a specific gameplay mechanic.

Task 4: Review Categorized Findings

Task 4 User Activity:

1. The user navigates through dashboard categories, including:
    - Positive/Negative Sentiment
    - Usability Issues
    - New Mechanic Feedback
    - Competitive Analysis
2. The user expands categories to view supporting player comments and discussion excerpts.
3. The user examines recurring themes generated by the classification system.
4. The user identifies major opportunities, concerns, and trends within the player community.

Outcome: The research and insights team can quickly synthesize large volumes of player feedback into actionable insights.

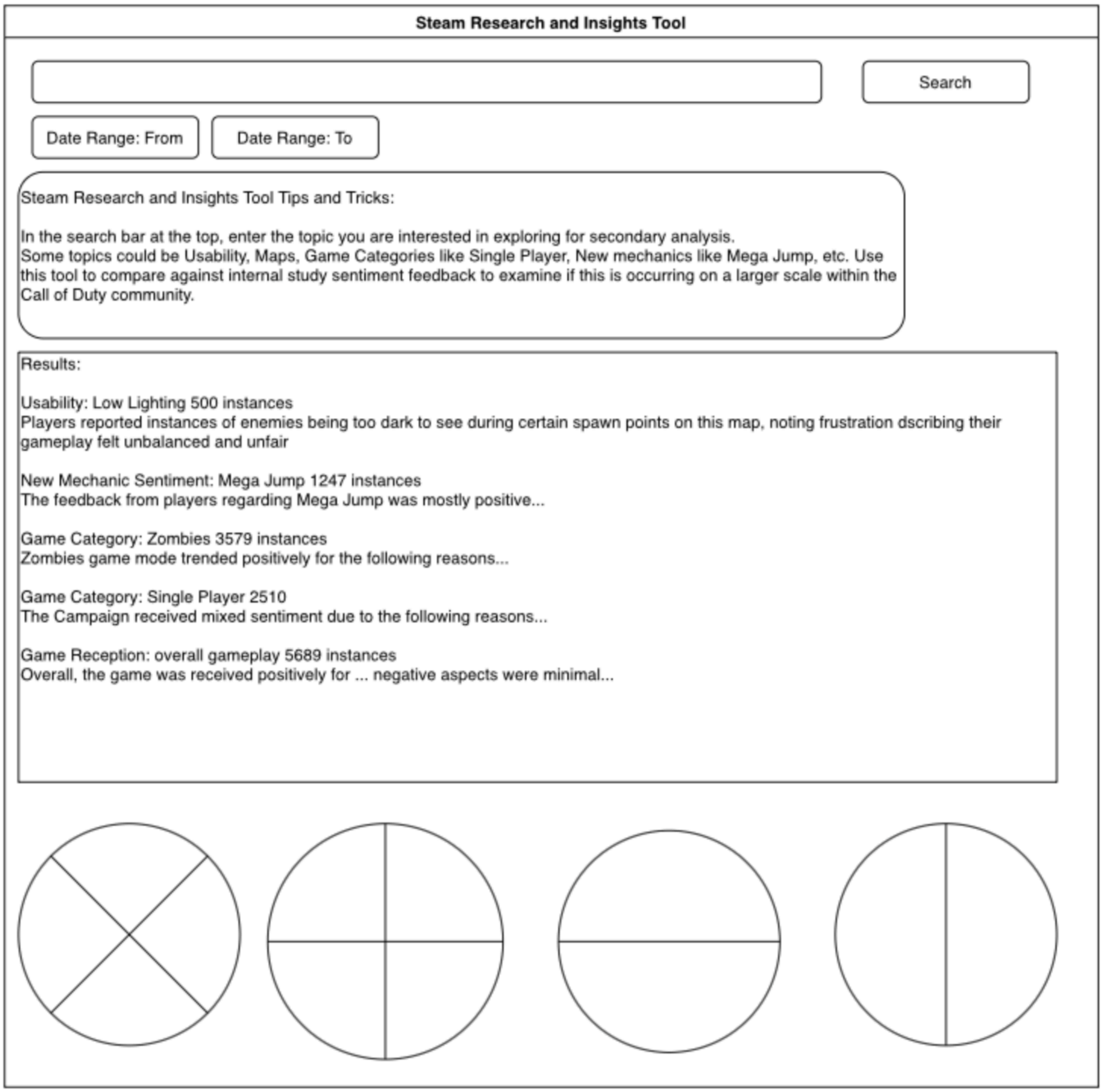
Task 5: Share Research Findings

Task 5 User Activity:

1. The user exports findings as a report, spreadsheet, or presentation-ready summary.
2. The user shares the report with Player Insights, Marketing, Product Management, and Development teams.
3. Stakeholders review the findings to inform:
    - Product decisions
    - Feature prioritization
    - Marketing messaging
    - Competitive positioning
    - Future research efforts
4. Teams use triangulated findings from both internal playtests and external Steam discussions to support decision-making.

Outcome: Insights from large-scale community discussions and internal research are communicated across teams to support evidence-based product and marketing decisions.

Diagram:



Technical "flow"

Research:

<u>Prof Harding:</u>

- Did some digging and found this was possible to do with Steam - Reddit is a bit more "gate-keepy" regarding scraping data or feedback from users
- Steam is more flexible when it comes to sharing data

<u>My Research:</u>

- Using AI web search (ChatGPT):
  - It is possible to pull player data from Steam
  - Recommends using FastAPI - easy to learn, but could use Flask, which is more beginner-friendly; however, not as robust as FastAPI
- Steam is "scraper" friendly:
  - https://dev.to/agenthustler/how-to-scrape-steam-in-2026-games-reviews-prices-and-player-data-2nc0
  - Use Beautiful Soup: https://www.geeksforgeeks.org/python/implementing-web-scraping-python-beautiful-soup/
- Steam data harvester resource: https://github.com/hpekkan/SteamDataHarvester

<u>Core functions:</u>

- Install Pandas
- Variables - define categories:
  - Usability
  - Sentiment
  - Mechanics

    categories = {
    "progression": ["xp", "leveling", "rank", "prestige"],
    "matchmaking": ["sbmm", "matchmaking", "lobby"],
    "maps": ["map", "maps"],
    "weapons": ["weapon", "gun", "shotgun", "rifle"],
    "performance": ["fps", "lag", "stutter", "crash"],
    "zombies": ["zombies", "undead"],
    "monetization": ["microtransaction", "bundle", "skin", "battle pass"]}
- Loops
- Requests
  - GET
  - POST
- Export to CSV
- Beautiful Soup (recommended by Steam Data Harvester and other sources listed above for web application)
- Pip install Flask
- OpenAI to summarize data
  - Categorize themes: from collections import Counter
  - def categorize_review(review):

```
review = review.lower()
found_categories = [ ]

for category, keywords in categories.items():
    for keyword in keywords:
        if keyword in review:
            found_categories.append(category)
            break
return found_categories
```

- Store data in a database for future use reviews.db
  - Allows faster searching

Program flow:

1. Data Collection Layer

Pull from:

- Steam Community Discussions
- Steam Reviews
- Steam Store Pages
- Web scraping

Output:

- Store posts, comments, reviews, dates, usernames, and game information in a database.

2. Data Processing Layer

This layer cleans and organizes the data.

Tasks:

- Remove spam
- Remove duplicate posts
- Extract keywords
- Categorize discussions

Output categories:

- New Mechanic Feedback
- Positive Sentiment
- Negative Sentiment
- Competitive Comparisons

3. Analysis Layer

Sentiment Analysis

Determine if comments are:

- Positive
- Neutral
- Negative

Final (self) assessment (1 pt)

- **After working through the spec, what was the biggest, most unexpected change you had to make from your sketch?**
    - I don't know if it was unexpected, but maybe something I wasn't considering was the AI summary or the AI grouping of themes. Looking into how to approach that in looking online, I think the most sense makes sense to look at how many times a topic comes up, so using a counting feature, because there are so many ways to define a problem or describe an issue in a video game, it is very subjective. That was kind of a blocker for me when I was thinking through this. I wasn't quite sure how I would go about defining categories or how to track them, since it is so subjective. I also hadn't considered storing the data somewhere for future use to measure sentiment over time, which in video game research can be very useful to have. It allows researchers and developers to see what has worked and what hasn't as new games are being made.
- **How confident do you feel that you can implement the spec as it's written right now?**
    - I feel somewhat confident. I think I need to put in a lot of work and meet with you regularly to make sure I understand what I am doing. It's a lot of small steps, so I need to make sure that I am not missing anything or forgetting a step. I'm not super confident regarding the web page portion, as I've never made a website before.

- **What is the biggest potential problem that you NEED to solve (or you'll fail)?**
    - I think the biggest potential problem is the sequence of steps and just how many steps are involved when executing the request and organizing the data. I think I need to be extra detail-oriented, which can be hard for me to make sure each step is constructed properly and then interacts correctly with each larger step so it flows correctly.
- **What parts are you least familiar with and might need my help with?**
    - I am least familiar with building a webpage application and the OpenAI component that I'm seeing in order to quickly summarize data themes. Same with using a web application tool, I have not used that before, so I might need your help regarding this, and maybe just an extra pair of eyes to make sure I don't forget a step.